CSci 135: M1 →

Ethics and Professional Practice

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Rodney Maniego Jr.
Department of Computer Science and Technology
Faculty of Computing * Visayas State University

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Course Outcome

Explain the concepts, principles, and professional context of software engineering.

Learning Outcome

Discuss the ethics and professional practice of software engineers.

Outline

1. SW Issues and its Impacts
2. ACM Code
3. Software Engineering Code

Ethics and Professional Practice

SW Issues and its Impacts

Popular SW Issues

1962

Atlas Agena (Mariner 1)

Omitted overbar (smooth average).



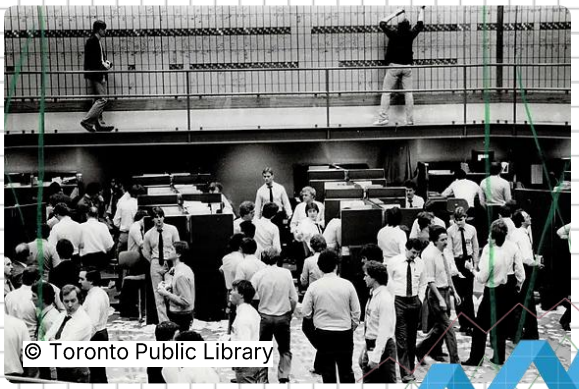
* An omitted mathematical symbol in the guidance software caused erroneous steering commands, leading to the destruction of the launch vehicle.

Popular SW Issues

1983

Vancouver Stock Exchange

Accumulated roundoff error.



* Repeated rounding errors in the stock index calculation accumulated over time, causing the reported index value to significantly deviate from its actual value.

Popular SW Issues

1985

Thorac-25 Radiation Therapy

Race conditions and Engineer overconfidence.

* inexperienced in time-critical systems.



* Software race conditions caused the radiation therapy machine to deliver massive radiation overdoses, resulting in patient injuries and deaths.

Read more: https://hackaday.com/2015/10/26/killed-by-a-machine-the-therac-25/

Popular SW Issues

1988

The Morris Worm

First internet-based computer worm.



* Robert Tappan Morris, a Cornell graduate student, released a worm, which rapidly spread across the Internet and disrupted thousands of computers.

Watch: https://www.youtube.com/watch?v=7BRTixIDzzE

Popular SW Issues

2004

MER Memory Anomaly

Incorrect configuration and flawed logic.



* A memory-management error caused the rover to repeatedly reboot, temporarily disrupting its operation.

Read more: https://llis.nasa.gov/lesson/1483

Popular SW Issues

2011

Mt. Gox Bitcoin Heist

Poor operational mechanisms and unencrypted hot wallets.



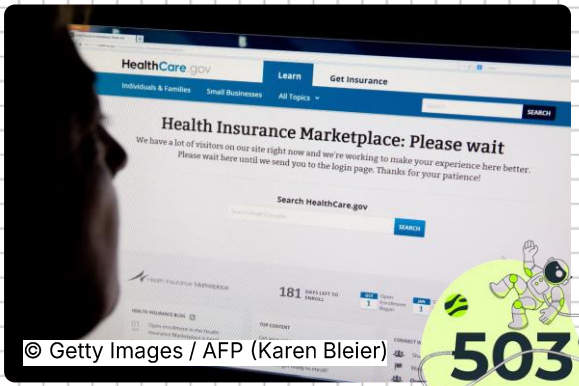
* A security vulnerability was exploited to steal ~2,609 BTC from the cryptocurrency exchange.

Read more: https://www.justice.gov/opa/pr/russian-nationals-charged-hacking-one-cryptocurrency-exchange-and-illicitly-operating-another0

Popular SW Issues

2013
HealthCare.gov Launch

Mission-critical software: Scalability and Usability issues.



© Getty Images / AFP (Karen Bleier)

* Poor system integration, performance issues, and inadequate testing caused widespread failures and made the website largely unusable at launch

Download: https://www.businessofgovernment.org/sites/default/files/Viewpoints%20Dr%20Gwanhoo%20Lee.pdf0

Popular SW Issues

2016

Gmail Mic Drop

April Fools' joke insult friends and lose jobs.



* An April Fools' feature accidentally replaced "Send & Archive," sending a Minion "mic drop" and muting email threads.

Read more: https://blog.google/products/gmail/introducing-gmail-mic-drop/

Popular SW Issues

2018

Facebook–Cambridge Analytica

3rd-party psychographic exploit on Facebook platform.



* Personal data from millions of Facebook users was harvested without proper consent and used for political profiling and targeted advertising.

Read more: https://en.wikipedia.org/wiki/Facebook%E2%80%93Cambridge_Analytica_data_scandal

Popular SW Issues

2024
CrowdStrike IT Outage

Out-of-bounds memory read at kernel-level.



* A faulty software update caused widespread Windows system crashes, disrupting airlines, banks, hospitals, and other services worldwide.

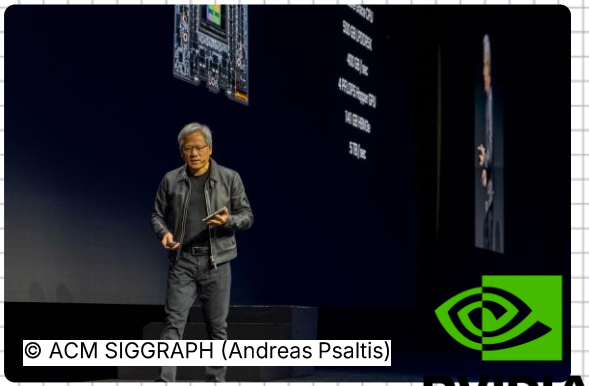
Watch: https://www.youtube.com/watch?v=4yDm6xNeYas

Popular SW Issues

2024

NVIDIA Cosmos Controversy

Unauthorized YouTube web scraping.



* NVIDIA was accused of using private/unlisted YouTube videos without permission to train its AI models.

Read more: https://www.404media.co/nvidia-ai-scraping-foundational-model-cosmos-project/

Ethics and Professional Practice

ACM Code

Association for Computing Machinery

It is a major ***international professional organization*** for computing and computer science.

Association for Computing Machinery

Code of Ethics and Professional Conduct

It serves as guide for the ethical conduct of all computing professionals.

* Covers current and aspiring practitioners, instructors, students, influencers, and anyone using computing technology.

Association for Computing Machinery

Code of Ethics and Professional Conduct

It reflect upon the wider impacts of their work, consistently supporting the public good.

* Must be adhered and promoted by ACM members.

General Ethical Principles



1.1 **Contribute to society** and to **human well-being**, acknowledging that all people are stakeholders in computing.

The results of efforts will respect **diversity**, meet **social needs**, be **used responsibly**, and be **broadly accessible**.

Read more: https://www.acm.org/code-of-ethics

General Ethical Principles



1.2 **Avoid harm.**

The “harm” means **negative consequences** that are significant and unjust.

Unintended harm must be **undo-ed** or **mitigated**; all harms must be **minimized** by following **general best practices**. If the leadership curtails mitigation, you are obliged to become the “**whistleblower**” to reduce potential harm.

Read more: https://www.acm.org/code-of-ethics

General Ethical Principles



1.3 **Be honest** and trustworthy.

Be honest in **competence** and **qualifications**.

Be transparent and provide full disclosure of **system capabilities**, **limitations**, and **potential problems**.

Should not misrepresent or speak on behalf an organization, unless authorized.

Read more: https://www.acm.org/code-of-ethics

General Ethical Principles



1.4 **Be fair** and take action not to discriminate.

Avoid creation of systems that promote **prejudicial discrimination** or disenfranchise people.

Harassment, bullying, and other forms of **abuse limits fair access**.

Technologies should be **inclusive** and **accessible**.

Read more: https://www.acm.org/code-of-ethics

General Ethical Principles



1.5 **Respect the work** required to produce new ideas, inventions, creative works, and computing artifacts.

"...**credit the creators** of ideas, inventions, work, and artifacts, and respect copyrights, patents, trade secrets, license agreements, and other methods of protecting authors' works."

Contribute to **free and open-source** community or the public domain.

Should **not claim private ownership** of any public resources.

Read more: https://www.acm.org/code-of-ethics

General Ethical Principles



1.6 **Respect privacy**.

Should be **well-informed** in forms of **data privacy**; including the rights and responsibilities associated with the **data collection** enabled by systems.

“Only the **minimum amount** of personal information necessary should be collected in a system.”

Personal information must not be used **without user consent**.

Read more: https://www.acm.org/code-of-ethics

General Ethical Principles



1.7 **Honor confidentiality**.

Trade secrets, client data, corporate financial and business information, and research data are **entrusted to employees** at varying degrees.

"...should **protect confidentiality** except in cases where there is evidence of a violation of law, of organizational regulations."

For information used as evidences, it should be **disclosed only to appropriate authorities**.

Read more: https://www.acm.org/code-of-ethics

Professional Responsibilities



2.1 Strive to **achieve high quality** in both the processes and products of professional work.

Insist **high-quality work** from oneself and others.

Respect all stakeholders by not degrading or exploiting them.

Avoid encouragement of **ignorance or neglect** resulting to poor quality.

Read more: https://www.acm.org/code-of-ethics

Professional Responsibilities



2.2 **Maintain high standards** of professional competence, conduct, and ethical practice.

Take **accountability** in maintaining professional competence.

It involves relevance **technical knowledge** and **awareness s**
to its social applicability.

It requires **independent pursuit to study**, attend technical trainings, and formal education.

Read more: https://www.acm.org/code-of-ethics

Professional Responsibilities



2.3 Know and **respect existing rules** pertaining to professional work.

Abide to organizational, local, national, and international laws, policies, and procedures unless there are inadequacy in moral basis or may cause recognizable harm.

Read more: https://www.acm.org/code-of-ethics

Professional Responsibilities



2.4 Accept and provide appropriate **professional review**.

Perform **peer and stakeholder review** at all stages of the development process.

Read more: https://www.acm.org/code-of-ethics

Professional Responsibilities



2.5 Give comprehensive and **thorough evaluations** of computer systems and their impacts, including analysis of possible risks.

Provide **credible evaluations** to stakeholders.

Strive **thorough and objective recommendations** of descriptions and alternatives.

Reassess to identify risks during software versions; report and **avoid deployment for major risks**.

Read more: https://www.acm.org/code-of-ethics

Professional Responsibilities



2.6 Perform work only in **areas of competence**.

Let go of assignments **outside of competence**.

Inform employer or client to pursue another with the **required expertise**.

Read more: https://www.acm.org/code-of-ethics

Professional Responsibilities



2.7 Foster **public awareness** and understanding of computing, related technologies, and their consequences.

Encourage **awareness and understanding** of computing in a clear, respectful, and welcoming.

Respectfully address inaccurate or **misleading information**.

Read more: https://www.acm.org/code-of-ethics

Professional Responsibilities



2.8 Access computing and communication resources **only when authorized** or when compelled by the public good.

Should **not access data and systems** if unauthorized or with exceptional circumstances.

Read more: https://www.acm.org/code-of-ethics

Professional Responsibilities



2.9 Design and implement systems that are robustly and **usably secure**.

Ensure **functional** suitability and usability.

Design against **accidental and intentional misuse**, modification, and other cybersecurity vulnerabilities.

If harm or misuse is not preventable, **do not implement the system**.

Read more: https://www.acm.org/code-of-ethics

Association for Computing Machinery

Code of Ethics and Professional Conduct



Contribute to society



Be honest and trustworthy



Respect the work of others



Honor confidentiality



Avoid harm



Be fair



Respect privacy



Professional Responsibilities



Leadership Responsibilities

Ethics and Professional Practice

Software Engineering Code

Software Engineering

Code of Ethics and Professional Practice

Software engineers shall commit themselves to making the analysis, specification, design, development, testing, and maintenance of software a ***beneficial and respected profession***.

Software Engineering

Code of Ethics and Professional Practice

Commitment to the public health, safety, and welfare.

* Ethical tensions are best addressed by consideration of fundamental principles, rather than blind reliance on regulations.

Principles



1. Public

Act consistently with the **public interest**.

Read more: https://ethics.acm.org/code-of-ethics/software-engineering-code/

Principles



2. Client and Employer

Act in the **best interests of their client and employer**, consistent to the public interest.

Read more: https://ethics.acm.org/code-of-ethics/software-engineering-code/

Principles



3. Product

Ensure that their products meet the **highest standards** possible.

Read more: https://ethics.acm.org/code-of-ethics/software-engineering-code/

Principles



4. Judgement

Maintain integrity in **professional judgments**.

Read more: https://ethics.acm.org/code-of-ethics/software-engineering-code/

Principles



5. Management

Leadership roles **embodies and promote** ethical management and software development.

Read more: https://ethics.acm.org/code-of-ethics/software-engineering-code/

Principles



6. Profession

Advance the **integrity and reputation** of the profession consistent with the public interest.

Read more: https://ethics.acm.org/code-of-ethics/software-engineering-code/

Principles



7. Colleagues

Be **fair to and supportive** to colleagues.

Read more: https://ethics.acm.org/code-of-ethics/software-engineering-code/

Principles



8. Self

Participate in **lifelong learning** regarding the **ethical practice** of their profession.

Read more: https://ethics.acm.org/code-of-ethics/software-engineering-code/

Software Engineering

Code of Ethics and Professional Practice



Public



Product



Manageme
nt



Colleagues



Client and
Employer



Judgement



Profession



Self

Summary

- ***Negligence of best practices*** or ***secrecy of harms*** can cause multi-pronged impacts.
- The ***ACM Code*** emphasizes on promoting public good, avoiding harm, upholding honesty, maintaining competence, respecting privacy, and practicing ethical computing.
- The ***ACM Code*** is broader in scope and addresses ethical principles with ***emphasis on societal impact***.

Summary

- The ***SE Code*** focuses in promoting public interest, ensuring product quality, upholding integrity, maintaining fairness, demonstrating competence, and exercising professional responsibility.
- The ***SE Code*** is narrower in scope but with detailed guidance on ***ethical practice*** and ***managerial duties***.

Scan me!

Evaluation Form

forms.gle/eUbH7NiBsvQGaPt27

Thank You!

References

- Association for Computing Machinery. (2018). ACM Code of Ethics and Professional Conduct. https://www.acm.org/code-of-ethics
- Gotterbarn, D., Miller, K., & Rogerson, S. (1997). Software engineering code of ethics. Communications of the ACM, 40(11), 110–118. https://doi.org/10.1145/265684.265699